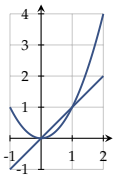
Form 24 Mar. 15 Dec. 2026 Resolution of f $(x) = g(x)$ *et* $f(x) < g(x)$

1. Warming · Flash [EQ] Solve 2 $x - 1 = x + 3$.

Rep. $x = 4$.

2. Heart · Graph [REG] Curves of $f(x) = x$ and $g(x) = x^2$:



Solve $f(x) = g(x)$ then $f(x) < g(x)$.

Rep. $x = x^2 \Leftrightarrow x(x-1) = 0 : x = 0$ or $x = 1$; $f(x) < g(x)$: the parabola is above the right for $x < 0$ or $x > 1$.

Intervals: $]-\infty;0[\cup]1;+\infty[$.

3. Heart + · Code reading [ALG] What's sent back `f(-1) == g(-1)` And? `f(0) == g(0)` ?

```
def f(x):
    return x*x
def g(x):
    returns 2*x + 3
```

Rep. True $(1 = -2 + 3)$ then False $(0 \neq 3)$: -1 is solution of $f(x) = g(x)$, not 0.

4. Back · Problem [MOD] Package A: 15,000 Ar fixed + 1,000 Ar per session. Package B: 2,500 Ar per session. From how many A sessions is it cheaper?

Rep. $15\,000 + 1\,000x < 2\,500x \Leftrightarrow x > 10$: from 11 sessions (10: 25 000 Ar equality).

5. Red thread · Flash [REP] Calculate the midpoint coordinates of the $[AB]$ with $A(-2;5)$ and $B(6;1)$.

Rep. $\left(\frac{-2+6}{2}; \frac{5+1}{2}\right) = (2;3)$.

Variable descriptive statistics

Form 25 game. 17 Dec. 2026 Mean and linearity of mean

1. Warming · Flash [STA] Average of 12, 15, 9, 14?

Rep. $\frac{50}{4} = 12{,}5$.

2. Heart · Method [STA] Series 4, 6, 8, 10, 12 averages 8. What is the average if we add 3 to If we multiply each value by 2?

Rep. Linearity: 8 $+ 3 = 11$; $2 \times 8 = 16$.

You don't have to do it again.

3. Heart + · True/false [STA] True or false? Justify. The average of two numbers $a < b$ is still strictly understood between a and b.

Rep. Right. $a < \frac{a+b}{2} < b$ since $b - a > 0$.

4. Back · Reading [LEC] In a circular diagram of 80 people, one sector represents 25%. What's his angle?

Rep. 20 people; 90 °.

5. Red thread · Flash [IVD] Calculate PGCD $(84\,;\,126)$ and deduce the irreducible form of $\frac{84}{126}$.

Rep. PGCD $= 42$; $\frac{84}{126} = \frac{2}{3}$.

Form 26 Mon. 4 Jan. 2027 Median and quartiles

1. Warming · Flash [STA] Range 7, 2, 9, 4, 5, 11, 3 then give the median.

Rep. 2 ; 3 ; 4 ; 5 ; 7 ; 9 ; 11 ; median 5 (range 4).

2. Heart · Method [STA] Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, median Q_3 and the interquartile gap.

Rep. $n = 8$: Q_1 row $\lceil 8/4 \rceil = 2$:5; median $\frac{8+9}{} = 8{,}5$; Q_3 row 6: 12; $Q_3 - Q_1 = 7$.

Convention: smaller value as at least 25 % (rep. 75 %) data are less than or equal to it.

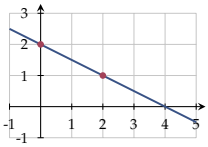
3. Heart + · True/false [STA] True or false? Justify. At least half of the values in a series are greater than or equal to its median.

Rep. True. By definition of median (at least 50 per cent of values on each side, including bounds).

4. Back · Flash [CMP] Store 0.6, $\frac{5}{8}$ 61% in increasing order.

Rep. $0{,}6 < 61\,\% = 0{,}61 < \frac{5}{8} = 0{,}625$.

5. Red thread · Graph → formula [REG] The right below represents an affine function f. Determine $f(x)$.



Rep. She's going through $(0;2)$ and $(2;1)$: slope $\frac{1-2}{2-0} = -\frac{1}{2}$, $f(x) = -\frac{1}{2}x + 2$.

Form 27 Mar. 5 Jan. 2027 Standard deviation; effect of adding or deleting a value

1. Warming · Flash [CAL] Calculate $(3-5)^2 + (7-5)^2$.

Rep. $4 + 4 = 8$.

2. Heart · Method [STA] Series 2 ; 4 ; 6 : mean, variance and standard deviation (value approximated to 10^{-2}).

Rep. Average 4 ; variance $\frac{4+0+4}{3} = \frac{8}{3}$; standard deviation $\sqrt{\frac{8}{3}} \approx 1{,}63$.

3. Heart + · True/false [STA] True or false? Justify. Add 5 to each value of a series does not change its Standard deviation.

Rep. True. All values and average are staggered by five: the deviations to the average are unchanged.

4. Back · Construction programme [chuckles] Let ABC a triangle, I the midpoint of $[AB]$ and D the symmetrical of C in relation to I. What is the nature of $ACBD$?

Rep. A parallelogram: its diagonals $[AB]$ and $[CD]$ have the same midpoint I.

5. Red thread · Flash [GEO] $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate AN.

Rep. Hales: $\frac{AN}{AC} = \frac{AM}{AB}$, therefore $AN = 10 \times \frac{3}{5} = 6$.

Form 28 7 Jan 2027 Comparison of series

1. Warming · Flash [STA] Range of 3, 8, 5, 12?

Rep. $12 - 3 = 9$.

2. Heart · Interpret [STA] Two series A and B have even average 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which one is More regular?

Rep. A : values are more tightened around the average (σ smaller).

3. Heart + · Interpret [STA] A : median 11, interquartile gap 2. B : median 13, interquartile gap 6. Compare the two series.

Rep. B is centered higher (median 13 > 11) but more dispersed (interquartile gap 6 > 2).

4. Back · Flash [PRB] We launch two balanced pieces. Likely to get two sides?